

Project Title:

Optimizing User, Group, and Role Management with Access Control and Workflows

Problem Statement:

In a project team consisting of a **Project Manager (Alice)** and a **Team Member (Bob)**, the absence of defined roles, access controls, and automated workflows caused confusion in task ownership and limited visibility into project progress.

This project implements a structured, role-based access and workflow system within the **ServiceNow platform**, enabling clarity, accountability, and automation in task management.

Objective:

To design and configure a **ServiceNow-based application** that:

- Defines clear roles and permissions for users and groups.
 - Implements **Role-Based Access Control (RBAC)**.
 - Automates task workflow transitions using **Flow Designer**.
 - Provides real-time visibility using **reports and dashboards**.
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Project Overview:

This application demonstrates how **ServiceNow configuration tools**—Users, Roles, ACLs, Flow Designer, and Analytics—can be integrated to create an efficient and secure project task management system.

Key Components:

- Role-based access for Managers and Members.
- Task management table for storing project tasks.
- Automated workflow that updates task status.
- Reports and dashboards for progress tracking.

Detailed Breakdown:**1. User, Group, and Role Creation:**

What it does: Creates distinct identities and assigns them specific access.

How it works:

- Users like Alice (Project Manager) and Bob (Team Member) were created.
- Groups were formed based on roles: Project Managers and Team Members.
- Custom roles such as `project_manager`, `team_member`, and `admin` were defined.
- Roles were assigned to users and linked to their respective groups, setting up the foundation for access control rules.

2. Custom Scoped Application: Project Task Tracker

What it does: Centralizes task management inside a dedicated app.

How it works:

- A scoped application called Project Task Tracker was created.
- A custom table Project Task was built with fields: Task Name, Description, Status, Assigned To, Due Date, Created By.
- This table serves as the main database to store and manage task-related information.

3. Role-Based Access Control (ACLs)

What it does: Restricts actions like creating, updating, reading, and deleting tasks based on user roles.

How it works:

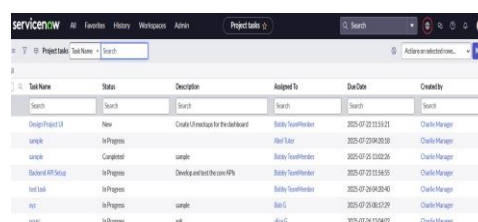
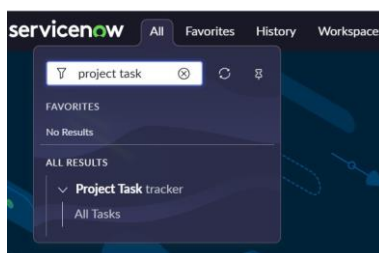
- ACLs (Access Control Rules) were set up for the Project Task table.
- Project Managers can create, update, delete any task.
- Team Members can only read and update tasks that are assigned to them.
- Roles were applied using “Requires Role” and Condition builder.

4. Application Navigation

What it does: Helps users quickly access the task list.

How it works:

- A custom Application Menu named Project Task Tracker was created.
- A module All Tasks was added under the menu.
- This module links to the list view of all project tasks.
- Role restrictions were used so that only users with correct roles can view this module.

A screenshot of the ServiceNow Project Task list view. The table has columns for Task Name, Status, Description, Assigned To, Due Date, and Created By. The first row is 'Design Project ID' with status 'New', assigned to 'Bobby Tasker', and due date '2025-07-22 13:35'. Other rows include 'sample' (In Progress), 'Backend API Setup' (In Progress), 'test task' (In Progress), 'xyz' (In Progress), and 'rxyz' (In Progress).

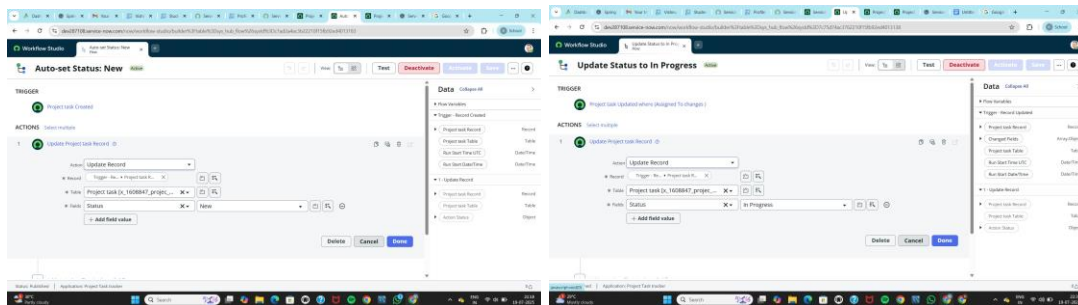
Task Name	Status	Description	Assigned To	Due Date	Created By
Design Project ID	New	Create Checksum for the dashboard	Bobby Tasker	2025-07-22 13:35	Charles Manager
sample	In Progress		Bob May	2025-07-23 09:28	Charles Manager
sample	Completed	sample	Bobby Tasker	2025-07-23 02:26	Charles Manager
Backend API Setup	In Progress	Develop and test the core API	Bobby Tasker	2025-07-23 13:35	Charles Manager
test task	In Progress		Bobby Tasker	2025-07-23 09:34	Charles Manager
xyz	In Progress	sample	Bob G	2025-07-23 08:29	Charles Manager
rxyz	In Progress	ask	Bob G	2025-07-23 09:02	Charles Manager

5. Workflow Automation Using Flow Designer

What it does: Reduces manual updates and keeps task progress accurate.

How it works:

- A flow was created using ServiceNow Flow Designer.
- Trigger: When a record in Project Task is updated.
- Condition: The Assigned To field is not empty
- Action: Automatically updates the Status field to In Progress.
- This helps to ensure that any time a task is assigned, its progress is tracked.



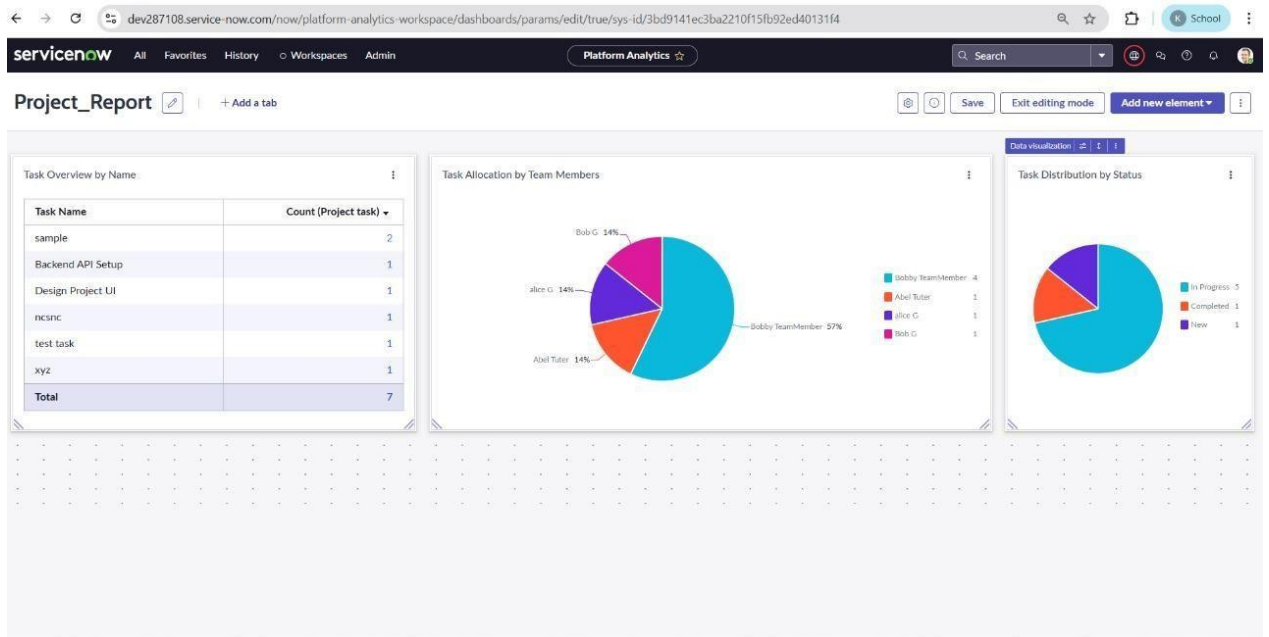
6. Dashboard & Reporting with Platform Analytics

What it does: Visualizes task progress and performance using charts and reports.

How it works:

- Multiple reports were created:
- Pivot Table grouped by Task Name
- Pie Chart grouped by Status (New, In Progress, Completed)
- Pie Chart grouped by Assigned Users
- These reports were added to a dashboard.

- The dashboard helps managers and team members monitor workload and status easily.



Outcome:

- Improved task visibility and accountability
- Controlled access based on user roles
- Automated workflow actions to reduce manual efforts
- Real-time monitoring of task distribution and progress

Why this is useful:

- Helps project managers and team members stay organized
- Prevents unauthorized task modifications

- Improves collaboration and transparency
- Supports scalable project tracking with clear metrics

Conclusion:

This project demonstrates how ServiceNow's built-in capabilities—**Users, Groups, Roles, ACLs, Flow Designer, and Analytics**—can be configured to create a secure and automated project workflow.

It serves as a reusable model for teams seeking structured access management and process automations

